

Relaxed Bipartite Matching with Structural Inertia for Efficient Diffusion

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Abstract

Bipartite matching has emerged as a highly efficient algorithmic paradigm for accelerating diffusion models, enabling massive sequence reduction with high parallel efficiency. However, we identify a theoretical limitation in this single-pass formulation: it imposes a strict *connectivity constraint* that forbids merging nodes within the same partition set, mathematically limiting the topological recall of semantic connections to 50%. This limitation manifests as *partition noise*, degrading the fidelity of dense generative tasks. To resolve this, we propose ReMatch (Relaxed Bipartite Matching), a training-free framework that treats sequence reduction as a constraint relaxation problem. ReMatch fundamentally reallocates the computational budget by synergizing two mechanisms: it employs cascaded, independent random partitions to relax the bipartite constraint and recover lost connections, while simultaneously exploiting the structural inertia of diffusion features to amortize the cost of these high-precision updates. Extensive experiments demonstrate that ReMatch establishes a superior Pareto frontier, effectively solving the under-clustering problem of standard bipartite approaches without incurring additional inference latency. Code is available at <https://github.com/mrazhou/ReMatch>.

CCS Concepts

• Computing methodologies → Machine learning algorithms.

Keywords

Diffusion Models, Token Merging, Efficient Inference, Bipartite Matching, Structural Inertia

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1 Introduction

The inference latency of diffusion probabilistic models [7, 11, 31] is severely constrained by the quadratic complexity of the self-attention mechanism, which models a fully connected graph of visual primitives. As image resolution increases, particularly in modern architectures [24] like FLUX [19] and Stable Diffusion [1], the computational cost grows prohibitively, hindering real-time applications. To alleviate this bottleneck, *dynamic graph coarsening* has been widely adopted [3, 20]. By aggregating redundant nodes based on semantic similarity, this process approximates the original dense graph with fewer vertices, thereby shrinking the effective sequence length while preserving essential visual information.

Prior approaches, most notably ToMe [2], formulate this coarsening process as a bipartite matching problem. Unlike global clustering algorithms (e.g., K-Means) or greedy search methods which incur prohibitive computational overhead ($O(N^2)$ iterative convergence or serial dependency), bipartite matching offers a highly parallelizable reduction step. Recent adaptations for diffusion, such as ToMeSD [3], have attempted to mitigate generation artifacts by introducing heuristics like strided sampling or local-random partitioning. However, these spatial strategies fail to address a fundamental topological bias inherent to the solver itself. We term this the *connectivity constraint*: in any single bipartite partition, connections between two nodes assigned to the same source set are structurally impossible, regardless of their semantic affinity. Existing heuristics merely redistribute these blind spots rather than eliminating them. This imposes a hard theoretical ceiling on matching recall (bounded at 50%), manifesting as *partition noise*, the erroneous separation of spatially coherent structures, which significantly degrades generation fidelity.

How can we break the topological limitations of the connectivity constraint to achieve high-fidelity generation, while maintaining the computational efficiency advantage of bipartite matching? To answer this, we propose ReMatch (**Relaxed Bipartite Matching with Structural Inertia**), a framework designed to bridge the gap between fast bipartite approximations and accurate global clustering. ReMatch

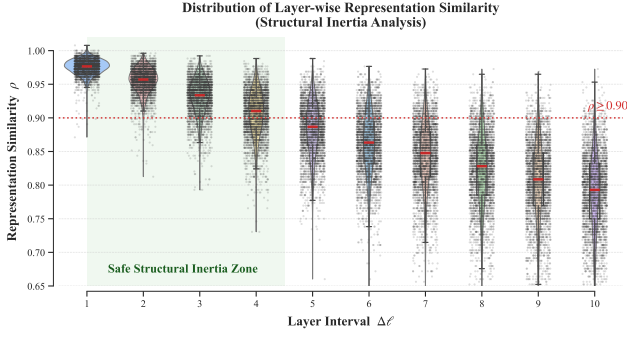


Figure 1: Statistical analysis of structural inertia on FLUX.1-dev (DiT). We visualize the distribution of Pearson correlation coefficients between bipartite matching matrices across varying layer intervals (Δl). The consistently high correlation at smaller intervals (median $\rho > 0.9$ for $\Delta l \leq 4$) indicates that merge topologies are stable across nearby transformer blocks.

addresses this by strategically shifting the computational focus, leveraging the synergy between two complementary mechanisms.

First, we introduce relaxed bipartite matching to break the topological recall limit. Instead of relying on a single constrained pass, we propose a cascaded matching mechanism using independent random partitions. Specifically, we apply a secondary partition with a distinct spatial configuration to the unmatched residuals of the first pass. This independence ensures that edges locked by the first partition are exposed in the second, effectively creating a second chance for nodes to form connections. This strategy mathematically lifts the recall ceiling and recovers fine-grained spatial details that would otherwise be discarded.

Second, to ensure this higher precision does not compromise latency, we exploit structural inertia. We observe that the optimal graph topology, specifically the set of ideal merge pairs, exhibits high temporal correlation across adjacent transformer blocks in diffusion backbones (as shown in Figure 1). ReMatch thus bundles the expensive metric computation and graph partitioning into sparse keyframes. Subsequent layers reuse this topology without re-computing the matching procedure. By shifting the computational budget from frequency (updating every layer) to quality (two-pass relaxation), ReMatch achieves a superior Pareto frontier between visual fidelity and inference speed.

Our contributions are as follows:

- We provide a graph-theoretic analysis of efficient attention approximations, identifying the connectivity constraint in single-pass bipartite matching as the cause of generation artifacts.
- We introduce ReMatch, a training-free framework that employs orthogonal relaxation to maximize topological recall and structural amortization to minimize latency.
- We demonstrate that ReMatch establishes a new state-of-the-art for training-free diffusion acceleration, consistently outperforming standard bipartite baselines in both fidelity and structural consistency.

2 Related Work

Efficient Diffusion Models. While diffusion probabilistic models have revolutionized generative tasks, their practical deployment is hindered by high inference latency, stemming from the iterative denoising process [29, 30] and the quadratic complexity of self-attention mechanisms [3, 22]. Acceleration techniques generally fall into two categories: training-based and training-free. Training-based methods typically employ knowledge distillation [33] or architecture search [13, 38] to compress models or reduce sampling steps [32, 36], often at the cost of retraining [21]. Conversely, training-free methods optimize inference without weight modification, utilizing strategies such as feature caching, sparse attention, token pruning, token merging, and KV caching [5, 12, 17, 26, 28]. Among them, sparse attention [41, 42] reduces attention cost by restricting token interactions, whereas token merging reduces the number of tokens before attention; these two directions are therefore complementary. A prominent subset of training-free methods further leverages temporal redundancy: DeepCache [22] reuses high-level U-Net features across steps, while others cache block-level or attention-layer outputs [5]. Further granular approaches like CAT Pruning and TokenCache exploit token-level redundancy across timesteps [8]. Our work sits within this training-free paradigm but distinctively targets *spatial* rather than temporal redundancy. We introduce *structural inertia*, positing that the topological structure of token merging remains highly stable across layers. This observation shares conceptual similarities with the pattern reuse in ToMA [20] but differs fundamentally in dimension. While ToMA exploits temporal redundancy by reusing matching patterns from the previous timestep for the current layer, our approach targets spatial depth redundancy, propagating the topological structure from preceding layers to subsequent ones within the same inference step. This enables amortized structural updates, effectively offering a spatial analogue to temporal feature caching.

Token Reduction and Merging. Dynamic graph coarsening has become a standard strategy to mitigate the quadratic cost of attention [15, 18, 34]. While token pruning effectively reduces computation, it risks discarding essential information [34]. Token merging offers a more preservation-friendly alternative by aggregating, rather than dropping, redundant nodes based on semantic similarity [3]. The adaptation of this method for Stable Diffusion, ToMeSD, has established a strong baseline for high-resolution generation using a bipartite source and destination formulation. Subsequent works have extended this via better GPU efficiency (ToMA) [20] and temporal reuse [26]. However, these methods predominantly rely on a single-pass bipartite matching formulation inherited from ToMe [2]. As we analyze, this heuristic imposes a strict *connectivity constraint*, rendering connections between nodes within the same partition structurally impossible. This limitation creates a theoretical recall ceiling of 50%, manifesting as *partition noise* that noticeably degrades fidelity in dense generative tasks.

Bipartite Matching in Vision. Bipartite matching provides a highly efficient mechanism for token reduction. Unlike global clustering algorithms such as K-means, which require computationally expensive iterative updates to converge [14, 23], bipartite matching enables single-pass assignment with high parallelizability. Standard

implementations like ToMeSD rely on a strict single-pass partition to assign merge pairs [3]. While heuristics like strided sampling or local-random partitioning attempt to mitigate the resulting artifacts, they function by merely spatially redistributing the topological blind spots rather than resolving the underlying connectivity constraint [18, 20]. We diverge from these single-pass heuristics by reformulating the problem as a cascaded relaxation. Instead of shifting partition boundaries, ReMatch employs an orthogonal matching strategy to mathematically approximate global clustering connectivity. This allows our framework to recover the topological recall lost by rigid bipartite constraints while preserving the linear complexity advantage through structural amortization.

3 The Bipartite Constraint

We formulate the token reduction process in diffusion transformers as a *dynamic graph coarsening* problem. Let $\mathcal{V}_l = \{x_1, \dots, x_N\}$ denote the set of tokens at layer l , where each token $x_i \in \mathbb{R}^d$. The objective is to identify a matching set $\mathcal{M}$ to merge redundant tokens, reducing the cardinality $|\mathcal{V}_l|$ by a ratio r .

Bipartite Partitioning. Current state-of-the-art methods (e.g., ToMeSD) adopt a divide-and-conquer strategy to achieve $O(N)$ complexity. They partition $\mathcal{V}_l$ into two disjoint sets: a Source set $\mathcal{S}$ and a Destination set $\mathcal{D}$, such that $\mathcal{V}_l = \mathcal{S} \cup \mathcal{D}$ and $\mathcal{S} \cap \mathcal{D} = \emptyset$. Edges are only permitted between $\mathcal{S}$ and $\mathcal{D}$, forming a bipartite graph $G_{bi} = (\mathcal{S} \cup \mathcal{D}, \mathcal{E}_{bi})$.

The Connectivity Constraint. While computationally efficient, this formulation imposes a strict topological constraint. For any pair of tokens $u, v \in \mathcal{S}$, the probability of forming a direct edge is strictly zero, regardless of their semantic similarity. Formally, let $\phi(u, v)$ be the ground-truth semantic affinity between tokens. The matching probability in a single-pass bipartite framework is:

$$P((u, v) \in \mathcal{M} \mid u, v \in \mathcal{S}) = 0, \quad \forall \phi(u, v) \rightarrow \infty. \quad (1)$$

We term this phenomenon *Partition Noise*. In dense prediction tasks like image synthesis, this constraint forces the model to discard highly correlated spatial features solely due to partitioning artifacts, leading to structural discontinuities in the generated latent space.

4 The ReMatch Framework

To resolve the conflict between computational efficiency and topological fidelity, we propose **ReMatch** (Relaxed Bipartite Matching with Amortized Structural Updates). ReMatch decouples the *quality* of the topology from the *frequency* of its update.

4.1 Relaxed Bipartite Matching (ReMatch)

We introduce a cascaded matching mechanism to relax the rigid constraint defined in Eq. 1. Instead of a single partition, ReMatch constructs the final topology via two orthogonal bipartite phases.

Phase I: Initial Coarsening. We first apply a standard bipartite matching. Let partitions be $\mathcal{S}_1, \mathcal{D}_1$. We solve for the partial matching $\mathcal{M}_1$ maximizing similarity scores within the bipartite constraint:

$$\mathcal{M}_1 = \arg \max_{\mathcal{M} \subset \mathcal{S}_1 \times \mathcal{D}_1} \sum_{(u,v) \in \mathcal{M}} \phi(u, v). \quad (2)$$

Upon determining $\mathcal{M}_1$, matched source tokens are merged into their corresponding destinations via weighted averaging. Consequently,

we obtain an updated residual token set $\mathcal{V}'$, which comprises the updated destination tokens (now carrying aggregated information) and the surviving unmatched source tokens.

Phase II: Orthogonal Relaxation. Standard bipartite token merging stops after the first pass. In contrast, ReMatch re-partitions the residual token set $\mathcal{V}'$ into $\mathcal{S}_2$ and $\mathcal{D}_2$ using an independent random bipartite partition. This second partition is orthogonal to the first one in the sense that token pairs blocked by the initial source–destination assignment can become valid candidates in the residual graph. We then compute a secondary matching $\mathcal{M}_2 \subset \mathcal{S}_2 \times \mathcal{D}_2$.

The final topology is obtained by composing the primary matching $\mathcal{M}_1$ with the residual matching $\mathcal{M}_2$, which maps the second-pass residual indices back to the original token space. Therefore, ReMatch covers a larger set of candidate merge connections than single-pass bipartite matching, while preserving the linear-time matching structure.

4.2 Amortized Structural Updates

Relaxed matching improves the quality of the merge topology, but recomputing the topology at every transformer block still introduces avoidable overhead. We further exploit the *structural inertia* of diffusion models: nearby transformer blocks tend to share similar merge topologies, i.e., which tokens should be merged remains stable over short layer intervals.

We introduce a **Keyframe Update Strategy**. We define a structural update period k .

- **Keyframe Layer** ($l \bmod k = 0$): We perform the full ReMatch procedure (Phase I + II) involving metric computation, sorting, and index generation. The resulting merge indices $\mathcal{I}_{key}$ are cached.
- **Inertial Layer** ($l \bmod k \neq 0$): We bypass metric computation and partitioning entirely. The layer simply inherits $\mathcal{I}_{key}$ to perform weighted pooling.

This effectively bundles the metric ϕ and the partition π into a unified structural state. By amortizing the cost of the expensive ReMatch over k layers, we achieve superior topological quality with negligible latency overhead.

Here, $\mathcal{T}^*$ denotes the cached merge topology, k is the structural update interval, and $\text{COMPOSE}(\pi_1, \pi_2)$ maps the second-pass residual matching back to the original token indices.

5 Theoretical Analysis

In this section, we analyze why ReMatch outperforms standard bipartite matching in inference-only settings, focusing on graph recall bounds and perturbation analysis.

5.1 Graph Recall Bound via Relaxation

Consider a set of tokens $\mathcal{V}$ and a set of ground-truth semantic edges $\mathcal{E}_{gt}$, where each edge indicates a token pair that is desirable to merge. We define the *candidate recall* $\mathcal{R}$ as the probability that an edge $e \in \mathcal{E}_{gt}$ is exposed as a valid candidate under the bipartite constraint. This quantity characterizes the accessibility of semantic connections before the final one-to-one matching selection, and is therefore a necessary but not sufficient condition for an edge to be selected.

Algorithm 1 Relaxed Bipartite Matching with Structural Inertia

Require: Token features $X_l \in \mathbb{R}^{N \times C}$ at layer l , reduction ratio r , inertia interval k .

Ensure: Merged tokens $X'_l \in \mathbb{R}^{N(1-r) \times C}$.

```

1: Static: Topology cache  $\mathcal{T}^*$  (initialized to  $\emptyset$ ).
2: {— Branch 1: Structural Inertia (Topology Reuse) —}
3: if  $l \not\equiv 0 \pmod{k}$  and  $\mathcal{T}^* \neq \emptyset$  then
4:    $\mathcal{T} \leftarrow \mathcal{T}^*$  {Reuse cached topology}
5: else
6:   {— Branch 2: Keyframe Update (Topology Re-calculation) —}
7:   {Phase I: Primary Bipartite Matching}
8:    $\mathcal{A}, \mathcal{B} \leftarrow \text{PARTITION}(X_l)$ 
9:    $\pi_1 \leftarrow \text{BIPARTITEMATCH}(\mathcal{A}, \mathcal{B} \mid \text{sim}(\mathcal{A}, \mathcal{B}))$ 
10:   $X_{\text{rem}} \leftarrow \text{WEIGHTEDMERGE}(X_l, \pi_1)$ 
11:  {Phase II: Relaxed Matching (The “ReMatch” Step)}
12:   $\mathcal{A}', \mathcal{B}' \leftarrow \text{RANDOMPARTITION}(X_{\text{rem}})$ 
    {Independent residual partition}
13:   $\pi_2 \leftarrow \text{BIPARTITEMATCH}(\mathcal{A}', \mathcal{B}' \mid \text{sim}(\mathcal{A}', \mathcal{B}'))$ 
14:  {Topology Consolidation}
15:   $\mathcal{T} \leftarrow \text{COMPOSE}(\pi_1, \pi_2)$ 
16:   $\mathcal{T}^* \leftarrow \mathcal{T}$  {Cache for subsequent layers}
17: end if
18: {— Execution: Feature Aggregation —}
19:  $X'_l \leftarrow \text{WEIGHTEDMERGE}(X_l, \mathcal{T})$ 
20: return  $X'_l$ 

```

THEOREM 5.1 (BIPARTITE RECALL LIMIT). *Under a random single-pass bipartite partition, the upper bound of the expected candidate recall for any edge $e \in \mathcal{E}_{\text{gt}}$ is 0.5.*

PROOF. Let $e = (u, v)$ be a target pair. In a bipartite matching framework, e is a valid candidate if and only if u and v are assigned to different sets (i.e., $u \in \mathcal{S}, v \in \mathcal{D}$ or vice versa). Assuming a uniform random partition where $P(u \in \mathcal{S}) = P(u \in \mathcal{D}) = 0.5$, the probability of separation is:

$$P(\text{valid}) = P(u \in \mathcal{S})P(v \in \mathcal{D}) + P(u \in \mathcal{D})P(v \in \mathcal{S}) = 0.25 + 0.25 = 0.5. \quad (3)$$

Thus, 50% of the valid semantic connections are strictly unreachable in a single pass. $\square$

THEOREM 5.2 (REMATCH RELAXATION). *Under two independent bipartite partitions, the expected candidate recall increases from 0.5 to 0.75.*

PROOF. Let A be the event that a target edge $e = (u, v)$ is exposed as a valid candidate in Phase I, and let B be the corresponding event in Phase II. From Theorem 5.1, $P(A) = 0.5$. When e is not exposed in Phase I, an independent residual partition gives it another chance to become a valid bipartite candidate. Under the independence assumption, $P(B \mid A^c) = 0.5$. Therefore, the probability that e is exposed in at least one phase is

$$P(A \cup B) = P(A) + P(A^c)P(B \mid A^c) = 0.5 + 0.5 \times 0.5 = 0.75. \quad (4)$$

Thus, the second partition recovers half of the semantic connections that are inaccessible in a single bipartite pass. $\square$

Remark. The above analysis focuses on candidate-edge accessibility rather than the optimality of the final selected matching. In practice, the one-to-one matching constraint still determines which exposed candidates are selected. Nevertheless, increasing candidate recall enlarges the feasible search space and reduces the chance that semantically similar tokens remain unreachable solely due to partition assignment.

Extension to Multiple Passes. With M independent bipartite partitions, the expected candidate recall becomes $\mathcal{R}_M = 1 - 2^{-M}$. Although more passes further reduce bipartite blind spots, the marginal gain diminishes rapidly and the residual token sets become increasingly sparse. As shown in Table 5, $M = 2$ achieves the best speed-fidelity trade-off in practice.

5.2 Perturbation Analysis: Topology vs. Metric

As our method is training-free, we analyze the error in terms of *inference feature deviation*. Let y_l be the output feature map of layer l . The deviation introduced by our method stems from two sources: topological error (γ) and metric approximation error (ϵ).

Topological Error (γ): This error arises when distinct semantic objects are erroneously merged due to Partition Noise (e.g., merging a “dog” token with a “background” token). This results in a *signal mixing* error. For a token x , if it is merged with a semantically unrelated token z , the output becomes $\frac{x+z}{2}$. The deviation is proportional to the feature distance $\|x - z\|$, which is typically large ($O(1)$) in the normalized feature space.

Metric Error (ϵ): This error occurs when we use a stale metric from layer $l-k$ to guide merging in layer l . Due to the smoothness of diffusion features [22, 35], the feature drift between adjacent blocks is small: $\|\phi_l - \phi_{l-k}\| \approx \delta$. Consequently, using a slightly stale metric results in only minor weighting discrepancies (e.g., merging x with neighbor x' instead of neighbor x'' , where $x' \approx x''$), resulting in a deviation of order $O(\delta)$.

Analysis. Topological errors introduce structural artifacts and boundary aliasing (high-frequency noise), which are detrimental to the dense prediction nature of diffusion models. In contrast, metric errors merely introduce negligible variance within semantically homogeneous regions. Therefore, our strategy represents a favorable computational trade-off: we invest the budget to minimize the dominant structural error γ (via ReMatch) while tolerating the minor approximation error ϵ (via Amortization).

6 Experiments

6.1 Experimental Setup

Architectures and Baselines. We conduct a comprehensive evaluation on two representative diffusion architectures: the U-Net-based Stable Diffusion v1.5 (SDv1.5) and the Transformer-based FLUX.1-dev. We compare ReMatch against state-of-the-art token reduction methods, specifically ToMeSD [3], SiTo [39] and ToMA [20], to benchmark efficacy across diverse backbones.

Benchmarks and Metrics. For quantitative fidelity and alignment evaluation, we utilize three standard benchmarks: ImageNet-1K [6], MS-COCO [4], and Aesthetic-4K [40]. For FLUX.1-dev, quantitative metrics are evaluated at 1024×1024 resolution unless otherwise specified. For SDv1.5, ImageNet experiments follow the standard

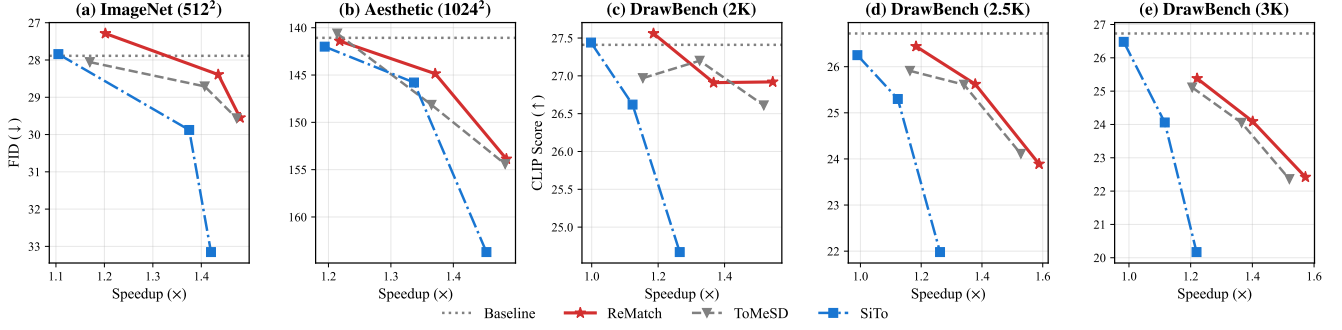


Figure 2: Fidelity-Efficiency Pareto Frontier across Scales. We visualize the trade-off between inference speedup (x-axis) and generation quality (y-axis) on five benchmarks of increasing difficulty: (a) Standard ImageNet generation on SDv1.5 (512^2), (b) High-fidelity Aesthetic-4K on FLUX.1-dev (1024^2), and (c)-(e) High-resolution DrawBench generation on FLUX.1-dev at 2048^2 , 3072×2048 , and 3072^2 . Note that for FID metrics in (a) and (b), the y-axis is inverted (lower is better), such that in all subplots, curves closer to the top-right corner indicate superior performance (higher speed and better quality). As resolution scales up (from left to right), baseline methods (ToMeSD, SiTo) exhibit severe performance degradation or structural collapse, whereas ReMatch maintains a robust advantage, consistently pushing the Pareto frontier outward.

Table 1: Quantitative comparison of token reduction methods on FLUX.1-dev. ReMatch (Ours) consistently achieves the best trade-off between inference speed and generation fidelity (FID, CLIP) across different reduction ratios.

Method	Ratio	Time (s) ↓	Speed ↑	COCO		ImageNet		Aesthetic-4K			
				FID ↓	CLIP ↑	FID ↓	CLIP ↑	FID ↓	CLIP ↑	IR ↑	Aesth ↑
FLUX.1-dev	-	18.36	-	32.31	30.94	38.14	30.48	141.07	33.78	1.34	6.97
ToMeSD	0.40	15.12	1.214×	30.11	31.14	37.04	30.54	140.61	33.69	1.27	6.72
SiTo	0.40	15.38	1.194×	29.23	31.15	36.56	30.54	142.00	33.77	1.29	6.81
ToMA	0.40	16.60	1.106×	30.48	31.00	36.75	30.58	139.81	33.48	1.32	6.95
ReMatch (Ours)	0.40	15.08	1.218×	28.77	31.16	36.50	30.54	141.41	33.77	1.29	6.81
ToMeSD	0.60	13.45	1.365×	35.10	31.12	39.14	30.56	148.15	33.41	1.11	6.47
SiTo	0.60	13.73	1.337×	36.41	30.93	43.16	30.36	145.79	33.54	1.17	6.60
ToMA	0.60	15.86	1.158×	31.98	31.00	38.03	30.66	142.58	33.35	1.26	6.94
ReMatch (Ours)	0.60	13.39	1.371×	33.23	31.15	38.71	30.58	144.86	33.59	1.16	6.55
ToMeSD	0.74	12.38	1.483×	46.45	30.93	46.29	30.66	154.41	33.34	0.87	6.05
SiTo	0.74	12.64	1.453×	68.52	29.96	61.61	29.84	163.71	32.77	0.79	6.10
ToMA	0.80	15.20	1.208×	35.69	30.94	39.89	30.71	148.08	33.33	1.19	6.96
ReMatch (Ours)	0.74	12.36	1.485×	46.22	30.88	45.68	30.59	153.87	33.27	0.88	6.04

512×512 setting. We report FID [10] to assess generation quality, along with CLIP Score [9, 25], ImageReward (IR) [37], and Aesthetic Score [40] to measure text-image alignment and perceptual aesthetics.

High-Resolution Scalability. To assess performance in extreme high-resolution scenarios, we employ the DyPE [16] extrapolation technique to scale generation from 2048×2048 up to 3072×3072 . Evaluations in this setting are conducted using prompts from the DrawBench [27] dataset, focusing on the model’s ability to maintain structural coherence under significant token expansion.

Implementation Details. Unless otherwise specified, ReMatch uses $M = 2$ relaxed matching passes and a structural inertia interval of $k = 4$ in all main experiments. Here, M denotes the number of

relaxed matching passes, k denotes the number of transformer blocks sharing the same merge topology, and the reduction ratio r denotes the fraction of removed tokens. All latency results are measured on a single NVIDIA RTX A6000 GPU and averaged over 100 runs. All methods are applied in a training-free manner without modifying model weights.

Iso-speed Configurations. For SDv1.5 experiments, Configs A–C denote iso-speed reduction settings used to compare generation quality at matched latency. With SDPA, Configs A/B use reduction ratios of 0.15/0.25/0.20 and 0.45/0.62/0.64 for ToMeSD/SiTo/ReMatch, respectively. Without SDPA, Configs A/B/C use 0.30/0.30/0.40, 0.60/0.65/0.55, and 0.70/0.70/0.65, respectively.

Table 2: Quantitative comparison on SDv1.5 (U-Net) on ImageNet using the Diffusers library with SDPA optimization. Configs A/B denote iso-speed reduction settings defined above.

Config	Method	Time (s) ↓	Speed ↑	FID ↓
-	SDv1.5	1.757	-	29.138
A	ToMeSD	1.594	1.102×	29.119
	SiTo	1.660	1.058×	28.968
	ReMatch	1.582	1.111×	28.839
B	ToMeSD	1.501	1.171×	29.518
	SiTo	1.584	1.109×	29.119
	ReMatch	1.468	1.197×	29.002

Table 3: Quantitative comparison on SDv1.5 on ImageNet using the original implementation without SDPA. Configs A–C denote iso-speed reduction settings defined above.

Config	Method	Time (s) ↓	Speed ↑	FID ↓
-	SDv1.5	3.018	-	27.888
A	ToMeSD	2.580	1.170×	28.057
	SiTo	2.732	1.105×	27.842
	ReMatch	2.511	1.202×	27.289
B	ToMeSD	2.144	1.408×	28.708
	SiTo	2.195	1.375×	29.877
	ReMatch	2.103	1.435×	28.391
C	ToMeSD	2.048	1.474×	29.563
	SiTo	2.125	1.420×	33.153
	ReMatch	2.039	1.480×	29.549

6.2 The Fidelity-Efficiency Pareto Frontier

We assess the efficacy of ReMatch by constructing the fidelity-efficiency Pareto frontier across two distinct diffusion architectures: the Transformer-based FLUX.1-dev and the U-Net-based Stable Diffusion v1.5 (SDv1.5). We demonstrate that ReMatch consistently yields a superior trade-off between inference latency and generation quality compared to state-of-the-art token reduction methods.

Pareto Superiority on DiT Architectures. As illustrated in Figure 2(b) and detailed in Table 1, ReMatch demonstrates exceptional robustness on the FLUX.1-dev model. In the moderate reduction regime (Ratio 0.4), ReMatch achieves a speedup of 1.218× while maintaining an FID of 36.50 on ImageNet, surpassing both ToMeSD (37.04) and SiTo (36.56). The performance gap widens significantly at aggressive compression rates. At a ratio of 0.60, ReMatch maintains an FID of 38.71, whereas the performance of baselines degrades noticeably (e.g., SiTo deteriorates to 43.16). Crucially, under the extreme setting of Ratio 0.74, while competing methods suffer from severe quality degradation—indicated by SiTo’s FID surging to 61.61—ReMatch preserves structural integrity with a significantly

lower FID of 45.68. This evidence suggests that the relaxed matching mechanism effectively identifies and preserves topologically critical tokens that rigid bipartite partitioning often discards.

Robustness to Hardware-Aware Optimizations (SDPA). A critical challenge for token reduction methods is their compatibility with existing compiler-level optimizations, such as PyTorch 2.0’s Scaled Dot Product Attention (SDPA). We investigate this on SDv1.5, reporting results in Table 2 (w/ SDPA) and Table 3 (w/o SDPA).

- *Standard Setting (w/o SDPA):* ReMatch achieves a quality-enhancing acceleration, where it accelerates inference while simultaneously improving image quality. Specifically, in Config A (Table 3), ReMatch attains a 1.20× speedup with an FID of 27.29, outperforming the original model’s FID of 27.89. This indicates that ReMatch acts as a form of structural regularization, filtering out redundant noise tokens.
- *Optimized Setting (w/ SDPA):* Even when the highly optimized SDPA kernel is enabled—which narrows the room for algorithmic acceleration—ReMatch retains its advantage. As shown in Table 2 (Config B), ReMatch delivers a 1.197× speedup with an FID of 29.00, consistently outperforming ToMeSD (1.171×, FID 29.52) and SiTo (1.109×, FID 29.12).

These results confirm that ReMatch is orthogonal to hardware-level optimizations, providing additive gains in practical deployment.

6.3 Scalability on High-Resolution Generation

While standard benchmarks demonstrate efficiency, the true stress test for token reduction lies in high-resolution generation ($> 2048^2$), where the quadratic growth of the attention matrix exacerbates the sparsity of semantic connections. We evaluate this scalability on the DrawBench dataset across resolutions from 2K to 3K (Table 4) and visualize the performance trajectories in Figure 2(c-e).

The Amplification of Topological Blind Spots. Our theoretical analysis posits that fixed-grid bipartite matching creates topological blind spots—connections within the same partition that are mathematically impossible to merge. As resolution scales, the spatial manifold expands, making these long-range dependencies increasingly critical for defining global structure. This hypothesis is empirically validated by the performance divergence observed in our experiments. At 2048^2 (Figure 2c), baseline methods remain competitive. However, a distinct *phase transition* occurs at 3072×2048 (Figure 2d), where the trajectories of ToMeSD and SiTo begin to decouple from the optimal frontier, indicating that single-pass matching becomes insufficient for capturing sparse semantic signals in high-dimensional spaces.

Resistance to Structural Collapse. The most striking result regarding quality emerges in the ultra-high-resolution regime (3072^2), where baseline methods suffer from catastrophic structural collapse under aggressive reduction ($r = 0.7$). As detailed in Table 4, SiTo exhibits a severe breakdown, with its CLIP score plummeting to 20.17 and ImageReward dropping to a negative value of -1.38 . This quantitative drop corresponds to the generation of incoherent, artifact-ridden images where semantic alignment is lost. In sharp contrast, ReMatch demonstrates exceptional resilience, maintaining a CLIP score of 22.42 and the highest ImageReward of -0.85 . By relaxing the strict bipartite constraint via multi-pass matching,

Table 4: Scalability Analysis on High-Resolution Generation (DrawBench). We evaluate performance across increasing resolutions ($2048^2 \rightarrow 3072^2$) using DyPE extrapolation. We observe that while baselines (ToMeSD, SiTo) suffer from performance degradation or structural collapse at higher resolutions (Right), ReMatch maintains superior robustness in both speedup consistency and metric retention.

Ratio	Method	Resolution: 2048×2048				Resolution: 3072×2048				Resolution: 3072×3072			
		Speed $\uparrow$	CLIP $\uparrow$	IR $\uparrow$	Aesth $\uparrow$	Speed $\uparrow$	CLIP $\uparrow$	IR $\uparrow$	Aesth $\uparrow$	Speed $\uparrow$	CLIP $\uparrow$	IR $\uparrow$	Aesth $\uparrow$
-	Baseline	1.000×	27.41	0.74	5.81	1.000×	26.72	0.51	5.61	1.000×	26.73	0.30	5.51
0.30	ToMeSD	1.154×	26.97	0.64	5.72	1.163×	25.91	0.26	5.35	1.205×	25.11	-0.07	5.15
	SiTo	0.998×	27.44	0.66	5.94	0.990×	26.25	0.37	5.57	0.983×	26.48	0.22	5.43
	ReMatch	1.186×	27.56	0.66	5.79	1.183×	26.44	0.37	5.45	1.221×	25.38	-0.01	5.20
0.50	ToMeSD	1.325×	27.20	0.47	5.52	1.341×	25.61	-0.06	4.99	1.365×	24.05	-0.46	4.90
	SiTo	1.123×	26.62	0.47	5.73	1.123×	25.30	0.11	5.44	1.116×	24.06	-0.29	5.26
	ReMatch	1.367×	26.91	0.50	5.59	1.378×	25.62	0.03	5.05	1.401×	24.09	-0.40	5.03
0.70	ToMeSD	1.518×	26.61	0.29	5.23	1.528×	24.11	-0.55	4.56	1.520×	22.36	-0.97	4.49
	SiTo	1.265×	24.67	-0.15	5.09	1.262×	21.98	-0.91	4.56	1.219×	20.17	-1.38	4.36
	ReMatch	1.545×	26.92	0.32	5.18	1.588×	23.89	-0.45	4.64	1.572×	22.42	-0.85	4.66

ReMatch effectively recovers the "blind" connections, preserving the semantic backbone of the image even when 70% of tokens are discarded.

Widening Efficiency Gap at Scale. Beyond quality, ReMatch exhibits a distinct asymptotic advantage in computational efficiency. As the token count N scales quadratically with resolution, the overhead of the matching algorithm itself becomes a non-negligible bottleneck. Observing Table 4 at 3072^2 , SiTo’s speedup stagnates at 1.22×, constrained by its compute-intensive similarity metrics. Even compared to ToMeSD, which achieves a respectable 1.52× speedup, ReMatch demonstrates superior efficiency (1.57×). The key differentiator is *structural inertia*. While ToMeSD incurs the overhead of reconstructing the bipartite graph at every timestep, ReMatch amortizes this cost by reusing the topological structure across intervals. This effectively decouples the matching overhead from the token scale, allowing ReMatch to maximize theoretical speedup gains in ultra-high-resolution scenarios without the redundant topological re-computation that limits baselines.

6.4 Ablation Studies and Mechanism Analysis

To deconstruct the performance gains of ReMatch, we conduct controlled experiments to isolate the contributions of our two core mechanisms: Relaxed Matching and Structural Inertia. Additionally, we benchmark ReMatch against K-Means clustering to evaluate the efficacy of local matching versus global optimization strategies in generative tasks.

Relaxed Matching and Structural Inertia. We jointly ablate the number of relaxed matching passes M and the structural inertia interval k in Table 5. The setting $M = 1, k = 1$ corresponds to the baseline without relaxed matching or topology reuse. Increasing M improves generation quality by exposing more candidate merge connections, while increasing k reduces the frequency of topology reconstruction. We use $M = 2, k = 4$ as the default setting because

it achieves a strong speed–fidelity trade-off. Further increasing M yields only marginal FID gains but introduces additional latency.

Table 5: Joint ablation of relaxed matching and structural inertia on FLUX.1-dev (COCO, Ratio=0.4). Each cell reports inference time / FID. The setting $M = 1, k = 1$ disables both relaxed matching and structural inertia, while $M = 2, k = 4$ is our default configuration.

M	Structural inertia interval k			
	$k = 1$	$k = 2$	$k = 3$	$k = 4$
1	15.64s/32.20	15.39s/31.82	15.12s/31.07	15.12s/29.59
2	15.79s/28.71	15.52s/28.63	15.12s/28.81	15.12s/28.77
3	16.12s/28.95	15.71s/29.41	15.53s/28.74	15.31s/28.69
4	16.60s/28.88	16.27s/28.99	15.95s/29.06	15.83s/28.37

Effect of Partition Strategy. We compare deterministic and random strategies for the second-pass residual partition in Table 6. The independent random partition performs better than the deterministic grid-shift strategy, indicating that random re-partitioning more effectively exposes token pairs blocked by the first bipartite assignment.

Table 6: Second-pass partition on COCO. FLUX.1-dev, $r = 0.4$.

Strategy	FID $\downarrow$
Deterministic grid-shift	29.65
Independent random partition	28.77

Compatibility with Sparse Attention. Token merging and sparse attention reduce inference cost from different perspectives: the former reduces the number of tokens, while the latter sparsifies

attention computation. We evaluate their compatibility in Table 7. ReMatch improves FID under comparable latency, and combining it with SparseAttention further reduces inference time, indicating that token merging and sparse attention are complementary.

Table 7: Compatibility with sparse attention on ImageNet. Experiments are conducted with FLUX.1-dev.

Method	Time (s) ↓	FID ↓
Baseline	18.36	38.14
SparseAttention [41] (topk = 0.5)	15.97	38.34
ReMatch ($r = 0.4$)	15.08	36.50
ReMatch + SparseAttention	14.58	38.28

Matching vs. Global Clustering. We also compare ReMatch with K-Means clustering in Table 8. K-Means directly clusters token features into $(1 - r)N$ centroids and requires iterative updates, which makes it less favorable under strict inference-time constraints. In contrast, ReMatch preserves the local matching structure while relaxing the bipartite constraint. As a result, ReMatch achieves lower latency and better FID than K-Means in this setting.

Table 8: Comparison with K-Means clustering on FLUX.1-dev (ImageNet, Ratio=0.6). K-Means clusters tokens into $(1 - r)N$ centroids with iterative updates, while ReMatch uses local bipartite matching with relaxed connectivity.

Method	Time (s) ↓	Speed ↑	FID ↓
Baseline (FLUX.1-dev)	18.36	1.000×	38.14
K-Means (iter=5)	13.90	1.321×	41.85
K-Means (iter=10)	14.20	1.293×	41.76
ReMatch (Ours)	13.45	1.365×	38.71

6.5 Qualitative Analysis

We provide visual evidence to corroborate our quantitative findings, highlighting the perceptual superiority of ReMatch in handling complex structures and textures.

Consistency at the Limit. Figure 3 demonstrates the robustness of ReMatch under extreme compression rates. At a 70% reduction ratio, the baseline method exhibits severe geometric distortion and semantic drift. In contrast, ReMatch maintains a coherent global structure and correct semantic alignment. This visual consistency aligns with the high ImageReward scores observed in Table 4, confirming that ReMatch effectively pushes the operational envelope of training-free acceleration.

Elimination of Partition Artifacts. As illustrated in Figure 4, baseline methods often introduce aliasing and discontinuities due to rigid partitioning. In the "modern building" example (Row 1), ToMeSD disrupts the continuous lines of the windows, whereas ReMatch preserves the geometric integrity. Similarly, in the "baby fennec" example (Row 4), SiTo fails to retain the high-frequency details of the fur, resulting in a blurred texture. ReMatch, by virtue of its

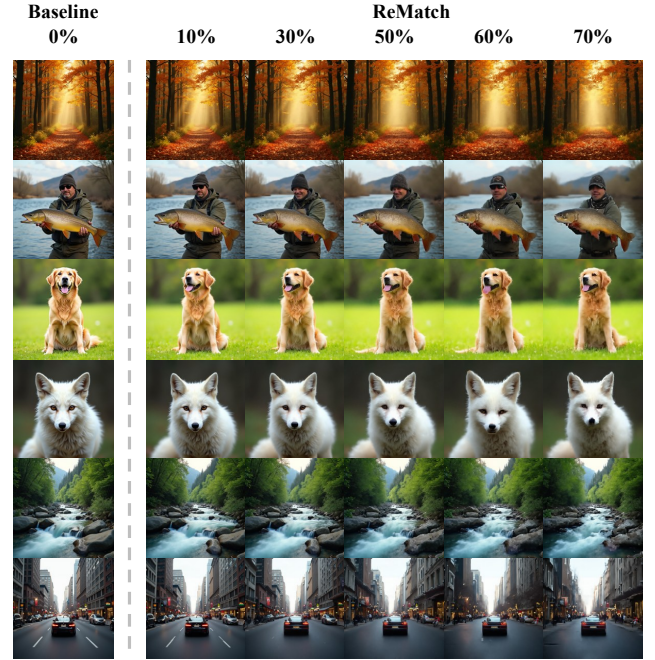


Figure 3: Impact of Reduction Ratios. While the baseline exhibits structural collapse at high ratios (e.g., distorted geometry at 70%), ReMatch preserves semantic integrity even under aggressive reduction.

relaxed connectivity, successfully reconstructs these fine-grained details, delivering photorealistic results that are perceptually indistinguishable from the original output.

Structural Integrity at Ultra-High Resolutions. We further investigate the robustness of token reduction methods under extreme resolution scaling (Figure 5). Using a fixed reduction ratio of $r = 0.3$, we generate images at resolutions ranging from 2048^2 to 3072^2 . The prompt specifically describes a single subject ("a ballerina"). As observed, baseline methods suffer from severe *semantic cardinality failure* as the resolution increases. Both ToMeSD and SiTo erroneously replicate the subject, generating multiple dancers or disjointed limbs (e.g., extra legs) in the 3072^2 setting. This phenomenon likely arises because their localized or rigid matching criteria fail to distinguish between repetitive texture patterns and distinct semantic objects over large spatial distances. In contrast, **ReMatch** maintains strict structural coherence, correctly rendering a single ballerina with anatomically accurate poses across all scales. This demonstrates that our relaxed matching mechanism effectively captures global semantic constraints, preventing the structural duplication artifacts that plague existing methods in ultra-high-resolution synthesis.

7 Conclusion

In this work, we presented **ReMatch**, a training-free framework designed to enhance the efficiency-fidelity trade-off in high-resolution diffusion models. By addressing the topological limitations of rigid

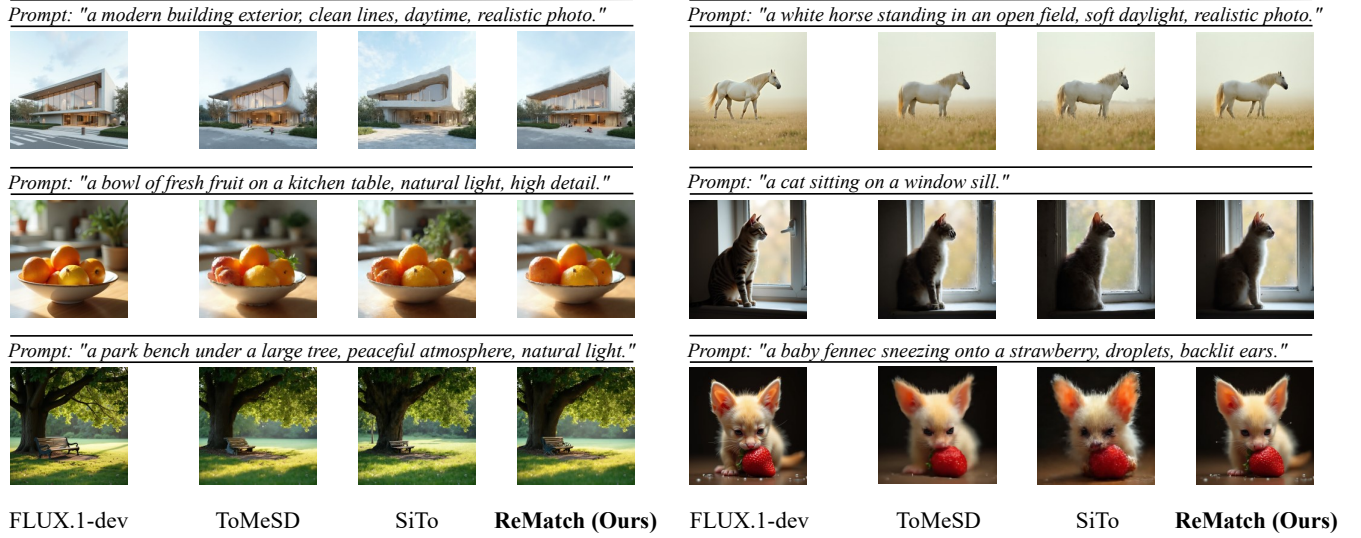


Figure 4: Visual Comparison. ReMatch reconstructs fine-grained details lost by other methods. Note the continuous architectural lines in the building (Row 1) and the realistic fur texture in the fox (Row 4), contrasting with the artifacts in ToMeSD and SiTo.



Figure 5: Qualitative comparison across resolutions ($2048^2 \rightarrow 3072^2$) with $r = 0.3$. As resolution scales up, baseline methods (ToMeSD, SiTo) exhibit catastrophic structural duplication artifacts, incorrectly generating multiple figures or redundant limbs. ReMatch (Ours) successfully preserves the semantic cardinality, rendering a single, coherent subject with correct anatomy even at the extreme 3072^2 resolution.

bipartite matching through *Relaxed Matching* and *Structural Inertia*, ReMatch achieves a favorable Pareto frontier across diverse architectures (U-Net and DiT). Notably, our experiments on SDv1.5 demonstrate that ReMatch accelerates inference while maintaining, and in some cases improving, generation quality compared to the full-token baseline. Furthermore, evaluations at ultra-high resolutions (3072^2) confirm that our approach effectively mitigates the structural degradation and semantic artifacts observed in existing methods, offering a robust, spatially-aware alternative to

global clustering strategies. Ultimately, ReMatch provides a scalable solution for high-fidelity generation on resource-constrained hardware.

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